

THE DRY-PACK CELLULOSE ALTERNATIVE

The dry-pack option offers an alternative to supercompetitive fiberglass insulation bidding or high-end spray foam installations.

by **BRUCE TORREY**

For many larger building and insulation contractors, getting into the new-construction market means entering the supercompetitive world of fiberglass bidding, or going for high-end spray foam installations. Neither of these makes sense for a small company whose main focus is on cellulose and weatherization. Many small weatherization contractors are quick to offer the benefits of cellulose insulation to the existing-home market. When it comes to new construction, these same contractors can be just as quick to pass on the benefits of cellulose.

The dry-pack option offers a great opportunity for small cellulose contractors to expand into the new-construction world. The main advantage is that start-up cost is minimal, because they can use the same equipment, crews, and techniques that they already have in place. The pricing strategies for dry-pack cellulose are also very favorable. The growth of the spray foam industry shows that many homeowners and builders are willing to pay more for a strategy that delivers better performance than fiberglass batts. Cellulose is perfectly positioned



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For smaller contractors, the dry-pack cellulose approach is a perfect method to test-drive new-construction contracting while supplementing their existing weatherization work. The main advantage is that start-up cost is minimal, because they can use the same equipment, crews, and techniques that they already have in place.

as being somewhat more expensive than fiberglass batts but substantially less expensive than spray foam, while its performance is arguably better than that of either.

The strengths of cellulose will help insulation contractors to make a convincing sales pitch:

- Unlike batts, cellulose completely fills the cavity, avoiding the problems caused by air convection inside the bay. The typical 5% void area seen in batt applications drops the performance of an R-19 batt to R-11, using weighted R-value calculations.
- The hygroscopic and air sealing characteristics of cellulose retard the flow of moisture-laden air into the cavity, and better manage any moisture that is already there. In new construction, it is not unusual for the cellulose to be drier than the adjacent wood framing.
- Dry-pack cellulose is also the perfect solution for insulating gutted cavities in rehab projects. Often these cavities have nonstandard depths that can be problematic if the standard-depth batt does not fill the cavity fully.

The Installation Process

The key to efficient installation is to use the right tools, and to follow an organized installation sequence. The dry cellulose is held in place with a polypropylene mesh netting system, stretched tight. First the netting is installed on the face of the studs, stretched hand tight, and stapled. The key to saving time is to install only a few staples. Initially, three or four staples per stud are sufficient to hold the material in place.

To stretch the netting supertight, fasten a series of side staples into the edge of the stud approximately $\frac{1}{8}$ inch in from the stud face. To expedite the process, I recommended using the Kihlberg pneumatic rapid-fire stapler, which fires at the incredible speed of 1,700 staples per minute. This lightweight tool offers the ultimate in one-handed, rapid-fire fastening. Again, the key is to use just enough staples on the stud face to hold the netting in place, and then use an additional six to ten staples on the stud edge, as needed, to tighten and stretch the netting material.

The next step is to apply roll-on or spray-on glue to weld the mesh netting to the interior face of the stud. This important step takes only a few minutes, but it is critical to keeping the cellulose off the stud face, where it would interfere with the wallboard application. The glue cures quickly, so there should be no downtime prior to blowing in the cellulose.

The installation process moves as fast as the blowing equipment allows. A slit in the netting somewhere in the middle of the bay allows the blowing tube to be inserted to the full height and depth of the cavity. Just as in the wall tube retrofit method, the bays are packed top and bottom first, working backward toward the middle insertion point. There's no need to turn the blower off; just move the hose to the next



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The "Insul-stud" system allows the blown in cellulose to act as a thermal break by filling around corners and inside headers.



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Rigid insulation board acts as part of the containment skin and provides a thermal break.



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The 2 x 4 studs built out with 2 x 2 nailers accommodate the blown-in cellulose.

bay. Contractor preferences vary when it comes to the best tube to insert. A flexible 2-inch wall tube or a length of metal or PVC pipe works well. For those averse to static shocks, we've even seen a cardboard mailing tube used successfully. A good feel for what a dense-packed bay should feel like, supplemented by an occasional open-handed slap on the netting, helps move the material into a consistent monolithic sealed bay.

The finished look of a well-packed bay provides a dramatic display for those who have never seen one. (For more on the meaning of dense-pack, see "Real-World Dense-Pack—Myth or Mystique?"). Building inspectors, homeowners, and other contractors all react with the same comments about how

full the bay looks, and how they wish they had used this technique on their own house. These glowing comments are immediately followed by negative comments from the always-skeptical drywall contractors, who are sure that the dense-packed cellulose will get in the way of their properly screwing the wallboard in place. Fortunately, a simple but ingenious tool called the belly roller solves the problem. This weighted roller looks at first glance like a standard paint roller. Working the roller over any bays that appear to be overfilled redistributes the cellulose perfectly flush with the stud face—ready for the most discriminating drywall installer.

Buildings with strapped cathedral ceilings typically don't need to be glued, since the netting can be installed below the 1 x 3 strapping and stapled to the rafter above, resulting in a tight enough configuration. Flat ceilings are insulated with loose-blown after the ceiling board and bypass air sealing work is completed.

Real-World Dense-Pack— Myth or Mystique?

The benefits of dense-pack cellulose have been the ongoing rallying cry and buzz phrase at countless insulation seminars and weatherization conferences. Critical to building shell performance, and promoted as a source of pride by many a contractor, the dense-pack mystique is unassailable. There's little question that dense-pack cellulose provides superior thermal performance and air leakage control, but how do we know that the dense-pack is really—well—dense?

Spot inspections I've completed on a wide range of contractors call into question the consistency of standards followed by and densities achieved by the typical contractor today. Despite the hoopla and the war stories about dense-packed cellulose blowing off wallboard, too many contractors are simply not installing cellulose at the densities expected. Part of the problem is that they don't know it.

One of the key components of a successful dense-pack installation is a powerful enough blower. Bill Hulstrunk, technical manager of National Fiber Cellulose, recommends a pressure of no less than 80 water column inches (80 inches WC, or 2.9 psi), measured at the blower outlet while blowing air and not material. These numbers seem to be appropriate, based on follow-up tests that Hulstrunk and I conducted with a calibrated cavity test box. Packing this test box of known weight and volume can reveal the net added weight of cellulose per cubic foot. Inspectors have yet to see a blowing machine that can

reach 80 inches WC and can't achieve a standard 3.5 lb of cellulose per cubic foot. The blowers that can't generate the necessary pressure can't pack the cellulose to the optimal density. Also, blowing equipment that blows with enough pressure to



Filling a calibrated test box to capacity ensures that the equipment is powerful enough for a successful dense pack installation and helps the crew to determine the best air/material ratio setting.



This dense-packed wall cavity was opened years after installation and the cellulose was still tight to the top plate.

dense pack the cellulose is easier and faster to work with. The cavity fills quickly and to the proper density.

There are a number of reasons for the lackluster densities. They include

1. old, undersized, or otherwise underperforming blower motors;
2. failure to maintain equipment, especially worn seals (when air can bypass worn seals, less pressure is available on the outlet side);
3. inadequate power due to low voltage or small-gauge wire;

4. a flawed air-material ratio (decreasing material feed and/or increasing blower air setting results in higher density); and
5. poor installer technique—including withdrawing the wall tube too quickly or not using a wall tube at all.

Consistent dense-pack is a readily obtainable and worthy goal, but ongoing testing indicates that it doesn't happen by accident. The following recommendations will greatly increase the likelihood of getting consistent dense-pack results:

1. Use an infrared thermal imager for quality control inspection.
2. Conduct equipment pressure-testing on a regular basis, supplemented by the appropriate ongoing maintenance program.
3. Train field crews with a calibrated test box to ensure that they have the right air-material ratio and installation technique.

Improvements in cellulose manufacture, advances in blowing equipment, and access to infrared cameras are moving the weatherization industry in a positive direction. For years we had to endure stories from homeowners who had remodeled their bathroom, only to find that their blown-in insulation had "settled." This led to the notion that cellulose settles more and more as time goes on. We now know that much of this cellulose was never dense-packed to begin with, and much of the "settling" had already taken place before the contractor's truck pulled out of the driveway. Fortunately, we now know how to avoid it.

Wall Options

The technique described above is compatible with any standard frame system. For colder climates, where a vapor barrier is deemed appropriate, a reinforced polyethylene material marketed as par/PAC is lip stitched tightly into place on the stud edge with a 1-inch crown pneumatic staple. The material serves as the netting for the insulation.

For the thermally adventurous, there are plenty of upgrade options to consider, including strategies for a good thermal break to slow conductive loss through the low-R-value wood frame members. A double wall of offset or staggered studs makes for a superinsulated R-30 cavity depth, perfect for dense-packing dry cellulose in the otherwise cold stud corners and headers. This wall system performs great, but is somewhat labor and material intensive.

A more familiar approach uses an interior rigid insulation board in conjunction with the blown-in cellulose (see middle photo, p. 31). When the rigid board is installed in sequence with the netting, it can eliminate some netting work by acting as the containment skin.

To keep with a more traditional jamb depth, a perpendicular hybrid wall system starts by framing a standard 2 x 4 exterior wall. A 2 x 2 wood strapping is then installed horizontally every 16 inches (see bottom photo, p. 31). The blown-in cellulose nicely fills the entire matrix and provides the thermal break while accommodating a standard 2 x 6 window and doorjamb.

The most practical system is marketed as the Insul-Stud (see top photo, p. 31). This hollow configuration of double 2 x 3s provides the ideal cavity for dense-pack, along with something not often seen—a perfectly straight



Consistent dense-pack such as that shown above is a readily obtainable and worthy goal, but ongoing testing indicates that it doesn't happen by accident.

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tion contractors and positions the weatherization folks to deliver a new-construction insulation strategy that really performs. Anyone who is familiar with the performance-challenged standards seen in new construction knows that this expertise is sorely needed.



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stud. This configuration is designed as a gusseted truss structural member and can be used for 16 inch or 24 inch on center framing. This system can be built to accommodate any wall depth—from a standard 5 1/2-inch depth to an R-40 wall. Not only does this design allow the cellulose to flow and fill around corners, but the open-ended stud allows the cellulose to pack in tight, filling the rough openings around window and doorjamb. This system, when specified from certified lumber and filled with recycled cellulose, is arguably the greenest wall option available to mainstream builders.

For the Small Contractor

Large-scale damp-spray cellulose strategies are ultimately the superior production-based approach for new construction, and many contractors are making that investment work. For smaller contractors, however, the dry-pack cellulose approach is a perfect method to test-drive new-construction contracting while supplementing their existing weatherization work. The lessons learned from dealing with insulation problems in existing homes and an in-depth understanding of air sealing, distinguish these weatherization contractors from strictly new home insula-

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Netting for densely packed dry-pack insulation is available through Hanes Industries. Go to www.hanesindustries.com.

The Kihlberg pneumatic stapler is available for order on the Web at www.kihlberg.com.

To make drywall contractors happy, purchase a weighted roller for installing densely packed dry-pack cellulose insulation at www.regalind.com.

To learn more about the par/PAC vapor barrier product, go to www.parpac.com.

To learn more about the Insul-Stud, go to www.energy-stud.com.